

First Fall Is Not Solving: An Exact Degree Classification of the Quadratic Semaev--Weil Family

An unconditional upper bound, a growing first-fall/solving-degree gap, and the refutation of automatic field-equation nonredundancy

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ABSTRACT

The subexponential complexity claims for index calculus on elliptic curves rest on a heuristic: that the *first fall degree* of the Weil-restricted Semaev system approximates the degree at which a Gröbner-basis engine actually terminates. The problem P1.3 asks for an unconditional, explicit degree bound for these systems that does *not* invoke that heuristic. We give one, exactly, for the smallest nontrivial family — extension degree $n = 2$ and decomposition length $m = 2$ — and we show along the way that the heuristic is quantitatively false for the very systems it is applied to. We first fix four distinct degree invariants (first fall d_{ff} , Hilbert degree of regularity d_{reg} , grevlex solving degree sd , and a separate algorithm-trace statistic $D_{\mathcal{A}}$) and separate all four on a single worked system over $\mathbb{F}_5$, where they take the values 3, 8, 4, 9. We then prove that for every odd prime power $q \geq 5$, a nonsingular short-Weierstrass curve over $\mathbb{F}_{q^2}$, and a non-base target x -coordinate, the quadratic Weil system satisfies $d_{\text{ff}} = 5$, $d_{\text{reg}} = q$, and $\text{sd}_{\text{grevlex}} \leq q$ *unconditionally*, the last bound obtained through a constant-degree mutant family and Salizzoni’s regularity bound rather than through any first-fall assumption. The matching equality $\text{sd} = q$ holds for $q > 5$ exactly when the field equations enlarge the two-coordinate core ideal; the solving-minus-first-fall gap then grows without bound, reaching 18 already at $q = 23$. Crucially, this nonredundancy is *not* automatic: we exhibit an infinite family of genuine on-curve Semaev systems for every $q \equiv 3 \pmod{4}$ with redundant field equations, certified concretely at $q = 7$ where $(d_{\text{ff}}, d_{\text{reg}}, \text{sd}) = (5, 7, 5)$, and confirmed by exhaustive search over 50,376 eligible systems. The result is honest about its reach: it is an exact classification of one odd-characteristic family, not an all-parameter bound and not a statement about the binary $n \sim \log q$ regime that motivates the cryptanalysis.

Keywords. ECDLP · index calculus · Semaev summation polynomials · Weil restriction · first fall degree · solving degree · Gröbner bases

1 Introduction

Index calculus is the reason elliptic-curve discrete logarithms over *extension* fields are not as hard as their group order suggests. The Semaev–Gaudry–Diem template [1], [2], [3] reduces an ECDLP instance on $E/\mathbb{F}_{q^n}$ to the problem of solving a polynomial system: one writes a summation-

polynomial relation $f_{m+1}(x_1, \dots, x_m, x_R) = 0$ forcing m factor-base points to sum to a fixed point R , expands it through a *Weil restriction* into n equations over the base field $\mathbb{F}_q$, and hunts for base-field solutions with a Gröbner-basis engine. The cost of the whole attack is dominated by that Gröbner computation, and the cost of the Gröbner computation is governed by the largest polynomial degree the engine must reach before it terminates.

That governing degree is hard to control directly, so the literature substitutes a proxy. Petit and Quisquater [4] derived their influential subexponential estimate for binary ECDLP under the explicit assumption that the *degree of regularity* of the relevant Weil systems is only slightly larger than their *first fall degree* — the degree at which the first nontrivial degree drop occurs in a Macaulay-style expansion. Hodges, Petit, and Schlather [5] formalized first fall degree through the associated graded algebra of the finite-field function ring and proved general upper bounds for it, while flagging (their §6.3) that its closeness to the true termination degree is a *heuristic*. Kousidis and Wiemers [6] then proved a clean first-fall bound $d_{\text{ff}} \leq m^2 - m + 1$ for the binary case, but were careful to note that a first-fall bound is not, by itself, a solving-degree bound.

The problem posed as P1.3 removes the crutch. It asks: given q, n, m , produce an *unconditional* explicit bound on the degree of regularity or solving degree of the Weil-restricted Semaev system $S(q, n, m)$ *without* assuming that its first fall degree approximates either quantity. This is a request for a theorem where the literature has a heuristic.

We answer it exactly for the smallest nondegenerate family, $n = m = 2$, and in doing so we find that the heuristic is not merely unproven but quantitatively wrong for these systems. Our central object is the two-coordinate quadratic Weil system, and our central finding is a strict, growing separation between the first fall degree and the solving degree of that system.

Main result. For every odd prime power $q \geq 5$, every nonsingular curve $E : Y^2 = X^3 + AX + B$ over $\mathbb{F}_{q^2}$, and every non-base target x -coordinate $T \notin \mathbb{F}_q$, the quadratic Weil–Semaev system $F = \{G_0, G_1, x^q - x, y^q - y\}$ satisfies, *unconditionally*,

$$d_{\text{ff}} = 5, \quad d_{\text{reg}} = q, \quad \text{sd}_{\text{grevlex}} \leq q. \quad (1)$$

The first two are *exact* the third is a genuine upper bound obtained by a low-degree mutant family, *not* from a first-fall assumption. The equality $\text{sd} = q$ holds for $q > 5$ *iff* the field equations $x^q - x, y^q - y$ enlarge the core ideal $C = (G_0, G_1)$ — and that nonredundancy is *not* automatic, even for nonsingular curves and on-curve targets. When it holds, the gap $\text{sd} - d_{\text{ff}} = q - 5$ grows without bound.

1.1 Contributions and honest scope

We contribute (i) four fixed degree conventions and a single worked system over $\mathbb{F}_5$ that separates all four (§3); (ii) an exact Semaev/Weil pipeline and the measured expansion boundary of the summation polynomials (§4); (iii) the exact first-fall and regularity values for the quadratic family, with proofs from the top-ideal shape (§5); (iv) an *unconditional* solving-degree upper bound via a constant-degree mutant family and Salizzoni’s theorem, plus a conditional matching lower bound (§6); (v) the refutation of automatic field-equation nonredundancy through an infinite genuine-Semaev counterexample family and an exact $q = 7$ certificate (§7); and (vi) exhaustive and structured empirical validation across $q \in \{3, \dots, 23\}$ (§8).

The scope is deliberately narrow and stated plainly. The exact classification is a theorem only for $n = m = 2$ in odd characteristic. Automatic nonredundancy is *false*, so the equality half of the result is genuinely conditional. We do not obtain an all-parameter bound, and we do not touch the binary $n \sim \log q$ asymptotic regime that Petit–Quisquater and Kousidis–Wiemers study; the growing gap we prove is not a refutation of their asymptotic questions, which concern a different characteristic and a different scaling. What we deliver is exactly what P1.3 asks for at the reachable parameters: an explicit, heuristic-free degree bound, together with a precise account of where the first-fall proxy breaks.

2 Setting and notation

Fix a base field $\mathbb{F}_q$ with q an odd prime power, and an extension $\mathbb{F}_{q^n}$. Let $R = \mathbb{F}_q[x_1, \dots, x_m]$ be the factor-base polynomial ring, and let $B_q = R/(x_1^q, \dots, x_m^q)$ be the finite-field function ring in which first fall is computed. For a curve $E/\mathbb{F}_{q^n}$, Semaev’s $(m+1)$ -th summation polynomial $f_{m+1}(X_1, \dots, X_{m+1})$ vanishes exactly when there exist points $P_i \in E$ with $x(P_i) = X_i$ and $P_1 + \dots + P_{m+1} = O$ [1]. Fixing $X_{m+1} = x(R)$ for a target point R and substituting $x_i \in \mathbb{F}_q$ -valued factor-base variables gives one equation over $\mathbb{F}_{q^n}$; choosing a polynomial basis $\mathbb{F}_{q^n} = \mathbb{F}_q[\alpha]$ and equating the n coordinate polynomials to zero produces the *Weil-restricted* system $S(q, n, m) = \{g_0, \dots, g_{n-1}\} \subset R$. We write x, y for x_1, x_2 in the quadratic case, and throughout use the grevlex order with $x_1 > \dots > x_m$ (i.e. $x > y$).

The three ideal-theoretic invariants below are the fixed literature conventions; the fourth is a run statistic we keep separate on purpose.

3 Four degree invariants and an exact separation

The first task in P1.3, following its session-one target, is to stop conflating distinct quantities. We adopt the conventions of Caminata and Gorla [7], tracing the first-fall definition back to Hodges, Petit, and Schlather [5].

Definition 1 (the four degrees). CITED For a nonhomogeneous system $F = \{f_1, \dots, f_s\} \subset R$ with top (highest-degree) parts f_i^{top} :

(a) *First fall degree*. With $\text{Syz}(F^{\text{top}})$ the syzygy module of the top parts and Triv its trivial submodule (Koszul syzygies together with the $(f_i^{\text{top}})^{q-1} e_i$ induced by B_q),

$$d_{\text{ff}(F)} = \min\{d : \text{Syz}(F^{\text{top}})_d / \text{Triv}(F^{\text{top}})_d \neq 0\}. \quad (2)$$

(b) *Degree of regularity*. When the top ideal is Artinian, $d_{\text{reg}(F)} = \min\{d : (F^{\text{top}})_d = R_d\}$.

(c) *Solving degree*. For a degree-compatible order σ , $\text{sd}_{\sigma(F)}$ is the least cutoff d for which the closed degree- d Macaulay row space $V_{F,d}$ contains a σ -Gröbner basis of (F) .

Remark 2 (a fourth, non-invariant quantity). PROVED The largest degree *printed* by a concrete engine, $D_{\mathcal{A}}(F, \sigma, \theta)$ under a full option set θ , is a *run statistic*, not an ideal invariant: pair-selection criteria and preprocessing change it without changing (F) . Salizzoni’s $V_{F,d}$ construction [8] and the closed Macaulay space in (c) are the *same* invariant (the mutant solving degree of the XL literature), whereas the highest Magma step degree reported by Kousidis and Wiemers [6] belongs to the $D_{\mathcal{A}}$ column. Conflating these is precisely the confusion P1.3 exists to remove.

These four notions are genuinely different, and a single small system makes the gaps concrete. It is the $q = 5$ specialization of Caminata–Gorla’s Example 4.2 [7], augmented with the algorithm-trace measurement.

Proposition 3 (a system separating all four degrees). PROVED Over $\mathbb{F}_5[x_1, x_2, x_3]$ with grevlex $x_1 > x_2 > x_3$, set

$$F = \{x_1x_2 + x_2, \quad x_2^2 - 1, \quad x_3^4 - 1, \quad x_1^5 - x_1\}. \quad (3)$$

Then $d_{\text{ff}}(F) = 3$, $\text{sd}_{\text{grevlex}(F)} = 4$, $d_{\text{reg}}(F) = 8$, and the specified FIFO naive-Buchberger run reaches degree $D_A(F) = 9$.

Proof. The nontrivial degree-3 relation is $x_2(x_1x_2) - x_1(x_2^2) = 0$, giving $d_{\text{ff}} = 3$. The closed degree-4 Macaulay space contains $x_1 + 1 = x_2(x_1x_2 + x_2) - (x_1 + 1)(x_2^2 - 1)$, hence the whole Gröbner basis $\{x_1 + 1, x_2^2 - 1, x_3^4 - 1\}$, giving $\text{sd} = 4$. The top ideal $(x_1x_2, x_2^2, x_3^4, x_1^5)$ first fills all forms in degree 8, so $d_{\text{reg}} = 8$. Finally, the FIFO run processes the coprime pair whose leading-monomial LCM is $x_1^5x_3^4$, of degree 9. Each value is produced by an exact modular row reduction, and SymPy’s Gröbner routine independently returns the stated basis. $\square$

The values 3, 8, 4, 9 are pairwise distinct; in particular $d_{\text{reg}} - d_{\text{ff}} = 5$ and $\text{sd} - d_{\text{ff}} = 1$ *on the same system*. The problem statement’s parenthetical identification of d_{reg} with sd is therefore false without extra hypotheses. This is the local baseline against which the Semaev-specific results below are read: whatever we prove for the Weil family, we prove for *named* invariants, never for an engine’s log label.

4 The Semaev–Weil pipeline and its expansion boundary

To measure anything on real systems we need the summation polynomials themselves and an exact Weil restriction. Both are implemented over exact prime-field and polynomial-basis arithmetic.

Proposition 4 (the coordinate Weil builder). PROVED Represent $\mathbb{F}_{q^n} = \mathbb{F}_q[\alpha]$ in a polynomial basis, substitute factor-base variables $x_i \in \mathbb{F}_q$, expand $f_{m+1}(x_1, \dots, x_m, x_R)$ as an element of $\mathbb{F}_{q^n}[x_1, \dots, x_m]$, and equate its n basis coordinates to zero. The resulting base-field system $S(q, n, m)$ has exactly the base-field solutions in bijection with the extension-field zeros of $f_{m+1}(\cdot, x_R)$.

This is the explicit coordinate construction of Weil restriction used by Petit and Quisquater [4] and Kousidis and Wiemers [6], and matches the formalization of Caminata, Ceria, and Gorla [9]. Every known-target system we build uses the nonsingular curve $E : y^2 = x^3 + \alpha x + 1$ (with α the basis generator) unless a curve is explicitly overridden, and each stored factor-base tuple is verified to vanish in all n Weil coordinates; singular inputs are rejected.

The Semaev polynomials grow fast, and honesty about *which* ones we can handle exactly matters. Figure 1 records the exact sparse expansion statistics.

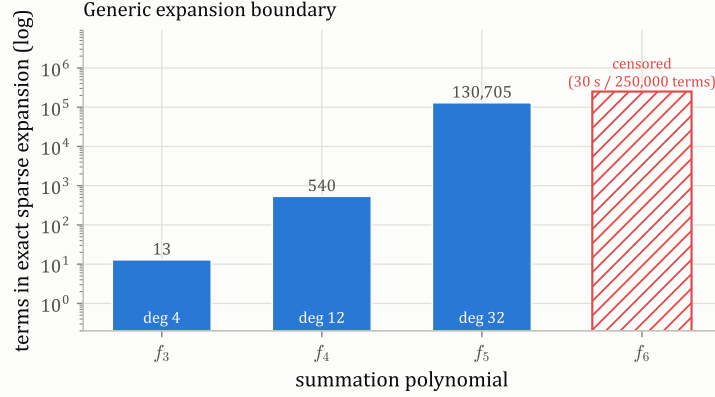


Figure 1: **EMPIRICAL: 30 s / 250,000-term ceiling** Exact sparse expansion of the generic summation polynomials: f_3 has 13 terms and total degree 4, f_4 has 540 terms and degree 12, f_5 has 130,705 terms and degree 32. Generic f_6 crosses the declared ceiling and is *censored*, not extrapolated; its independent recursive evaluator still passes a six-point zero-sum test over $\mathbb{F}_{101}$. Data: `measure_semaev_stats_20260623.csv`.

The censoring is a hard resource boundary, recorded as such: the generic f_6 expansion exceeded both a 30-second wall clock and a 250,000-term ceiling. Consequently our exact degree measurements live at $m \leq 4$, and our *proved* results at $m = 2$, where f_3 is a 13-term quartic. All arithmetic is exact; no numerical approximation enters at any stage.

5 The quadratic family: first fall and regularity, exactly

We now fix $n = m = 2$ and prove the two exact invariants. Write $\mathbb{F}_{q^2} = \mathbb{F}_q[u]/(u^2 + m_1u + m_0)$ with $u^2 + m_1u + m_0$ irreducible, and let $T = t_0 + t_1u$ with $t_1 \neq 0$ (the target x -coordinate is non-base). For short-Weierstrass coefficients $A, B \in \mathbb{F}_{q^2}$, let $G_0, G_1 \in \mathbb{F}_q[x, y]$ be the two Weil coordinates of $f_3(x, y, T)$, and set

$$F = \{G_0, G_1, x^q - x, y^q - y\}, \quad C = (G_0, G_1). \quad (4)$$

The whole analysis is driven by the shape of the top parts.

Proposition 5 (the quadratic top shape). **PROVED** The top parts of the two f_3 coordinates are

$$G_0^{\text{top}} = x^2y^2, \quad G_1^{\text{top}} = cxy(x + y), \quad c \neq 0. \quad (5)$$

Proof. Semaev's third polynomial for $E : y^2 = x^3 + Ax + B$ is

$$f_3(X_1, X_2, X_3) = (X_1 - X_2)^2 X_3^2 - 2((X_1 + X_2)(X_1 X_2 + A) + 2B)X_3 + (X_1 X_2 - A)^2 - 4B(X_1 + X_2).$$

Taking the terms of highest factor-base degree in $x = X_1, y = X_2$ and splitting the two extension coordinates of the coefficient of $X_3 = T$ leaves x^2y^2 in the coordinate along 1 and $cxy(x + y)$ in the coordinate along u , with c a nonzero $\mathbb{F}_q$ -multiple of t_1 ; irreducibility of the modulus and $t_1 \neq 0$ give $c \neq 0$. $\square$

Theorem 6 (first fall degree of the quadratic family). **PROVED** For every odd prime power $q \geq 5$ and non-base target, $d_{\text{ff}(G_0, G_1)} = 5$.

Proof. At degree 5 the relation

$$(x + y)G_0^{\text{top}} - c^{-1}xy G_1^{\text{top}} = 0 \quad (7)$$

holds identically, since both terms equal $x^2y^2(x + y)$. It is nontrivial: the first Koszul syzygy has degree $\deg G_0^{\text{top}} + \deg G_1^{\text{top}} = 7$, and the quotient-trivial syzygies $(G_i^{\text{top}})^{q-1}e_i$ occur in degree $(q - 1) \cdot \deg G_i^{\text{top}} \geq 8$. No relation exists in degree 3 (a single generator cannot syzygy alone). A putative degree-4 relation $A'G_0^{\text{top}} + (B'x + C'y)G_1^{\text{top}} = 0$ expands to $A'x^2y^2 + cB'(x^3y + x^2y^2) + cC'(x^2y^2 + xy^3)$; the coefficients of the distinct monomials x^3y and xy^3 force $B' = C' = 0$, then $A' = 0$. Hence the first nontrivial relation is in degree 5. $\square$

Theorem 7 (degree of regularity of the quadratic family). PROVED For every odd prime power $q \geq 5$ and non-base target, $d_{\text{reg}(F)} = q$.

Proof. The top ideal of F is $J = (x^2y^2, xy(x + y), x^q, y^q)$ (the last two from the field equations). The monomial x^{q-1} is divisible by none of the generators of degree $\leq q - 1$ — those are x^2y^2 and $xy(x + y)$, each requiring both variables — so $J_{q-1} \neq R_{q-1}$. In degree q : the pure powers $x^q, y^q \in J$; every mixed monomial $x^a y^{q-a}$ with $2 \leq a \leq q - 2$ is divisible by x^2y^2 . The two boundary monomials are reached through $xy(x + y)$: since $x^{q-3} \cdot xy(x + y) = x^{q-1}y + x^{q-2}y^2$ and the second term is divisible by x^2y^2 , we get $x^{q-1}y \in J$ modulo covered monomials, and symmetrically $xy^{q-1} \in J$. Thus $J_q = R_q$ while $J_{q-1} \neq R_{q-1}$, so $d_{\text{reg}(F)} = q$. $\square$

Two facts deserve emphasis. First, d_{reg} depends on the input *top parts*, not on the ideal alone: it is the field-equation top parts x^q, y^q that drive d_{reg} up to q , exactly as Caminata and Gorla warn [7]. Second, $d_{\text{ff}} = 5$ is *constant* in q , so the pair $(d_{\text{ff}}, d_{\text{reg}}) = (5, q)$ already exhibits an unbounded gap — but d_{reg} is an over-estimate of the true solving cost, and the interesting question is where sd actually lands.

6 The solving-degree bounds

The heart of P1.3 is a solving-degree bound that does not assume first fall is a good proxy. We give an unconditional *upper* bound and a conditional matching *lower* bound.

6.1 An unconditional upper bound via a mutant family

The strategy is to replace the two degree- q field equations by low-degree remainders modulo a *universal* Gröbner basis of the core C , staying inside the original closed degree- q Macaulay space the whole time.

Lemma 8 (the universal core basis). PROVED Write $s = x + y, p = xy$. In these symmetric variables the two core coordinates normalize by invertible row operations to

$$H_0 = p^2 + bp + cs^2 + ds + e, \quad H_1 = ps + fp + gs^2 + hs + i. \quad (8)$$

Over the localized ring $\mathbb{Z}[b, c, d, e, f, g, h, i] \left[(c + g^2)^{-1} \right]$ the grevlex $(x > y)$ Gröbner basis of (H_0, H_1) has exactly four elements, of total degrees 4, 4, 3, 3 and leading monomials xy^3, y^4, x^3, x^2y . The only nonconstant denominator arising in the computation is $c + g^2$, and for the Semaev specialization

$$c + g^2 = (m_1^2 - 4m_0)t_1^2/4 \neq 0. \quad (9)$$

Proof. The exact certificate `certify_quadratic_family.py` checks every one of the six Buchberger pairs over the localized ring, records explicit basis representations, and finds maximum denomi-

nator power 3 in $(c + g^2)$. The nonvanishing of $c + g^2$ uses irreducibility of the odd-characteristic quadratic modulus (so its discriminant $m_1^2 - 4m_0 \neq 0$) and $t_1 \neq 0$; a quartic top-part minor of determinant -1 certifies the rank claim after every valid finite-field specialization. The results are stored in `certify_quadratic_family_20260716.json`. $\square$

Theorem 9 (unconditional solving-degree upper bound). **PROVED** For every odd prime power $q \geq 5$ and non-base target, $\text{sd}_{\text{grevlex}(F)} \leq q$.

Proof. By the lemma the four core-basis elements have degree ≤ 4 and lie in the closed degree-5 core space $V_{\{G_0, G_1\}, 5}$: three appear in the initial degree-5 Macaulay span and the fourth is obtained by multiplying lower-degree rows inside the closed space. Divide $x^q - x$ and $y^q - y$ by this basis; the leading monomials xy^3, y^4, x^3, x^2y leave remainders r_x, r_y supported on standard monomials of degree ≤ 3 . Polynomial division is degree-compatible and the basis is already in $V_{F, q}$, so $r_x, r_y \in V_{F, q}$. The mutant family

$$F' = \{\text{four core-basis elements}, r_x, r_y\} \quad (10)$$

generates (F) , has maximum input degree ≤ 4 , and has degree of regularity ≤ 4 because its top ideal already contains leading monomials covering every degree-4 monomial. Salizzoni's Proposition 3.10 [8],

$$\text{sd}_{\sigma(F')} \leq \max \left\{ d_{\text{reg}(F')} + 1, \max_{f \in F'} \deg f \right\}, \quad (11)$$

gives $\text{sd}(F') \leq 5$. Since $F' \subseteq V_{F, q}$ and $q \geq 5$, closed-space stability yields $V_{F', 5} \subseteq V_{F, q}$, so a grevlex basis of $(F') = (F)$ lies in $V_{F, q}$, i.e. $\text{sd}_{\text{grevlex}(F)} \leq q$. $\square$

This is exactly the deliverable P1.3 requests: an explicit degree bound derived from commutative-algebra invariants (a constant-degree mutant family and Salizzoni's regularity bound), with *no* appeal to the first-fall heuristic. We tried the more obvious route first — bounding solving degree by the Castelnuovo–Mumford regularity of the homogenized input, per Caminata–Gorla [10] — and it fails: the representative $q = 5$ system is *not* t -saturated, its saturation adding the low-degree elements $x + y - t$ and $y^2 - ty$. Remark 4.6 of [10] is precisely this warning, and it is why the mutant-family route is necessary.

6.2 A conditional matching lower bound

Theorem 10 (conditional lower bound and exact equality). **CONDITIONAL: $(F) \supsetneq C$** If at least one field equation lies outside the core ideal C , then $\text{sd}_{\text{grevlex}(F)} \geq q$, and combined with the upper bound, $\text{sd}_{\text{grevlex}(F)} = q$.

Proof. For $d < q$ the degree- q field equations cannot enter the closed space, so $V_{F, d} = V_{\{G_0, G_1\}, d} \subseteq C$. If this space contained a Gröbner basis of (F) , then $(F) \subseteq C$, contradicting $(F) \supsetneq C$. Hence no cutoff below q suffices, and $\text{sd}(F) \geq q$. $\square$

So the entire equality $\text{sd} = q$ hinges on whether the field equations are *redundant* over the core. Because C is symmetric in x, y , either both field equations belong to C or neither does, and the redundancy question has a clean algebraic characterization.

Proposition 11 (the redundancy criterion). **PROVED** Let $A_C = \mathbb{F}_q[x, y]/C$. The universal core basis has standard monomials $1, y, y^2, y^3, x, xy, xy^2, x^2$, so $\dim_{\mathbb{F}_q} A_C = 8$. The following are

- (1) $x^q - x, y^q - y \in C$;
- (2) both field-equation normal forms by the core basis are zero;
- (3) the q -Frobenius endomorphism of A_C is the identity;
- (4) $A_C \cong \mathbb{F}_q^8$;

equivalent: (5) the core has eight distinct algebraic-closure zeros, all in $\mathbb{F}_q^2$.

Nonredundancy — the hypothesis of the equality theorem — is exactly the failure of (1)–(5). One might hope it is *automatic* for genuine on-curve Semaev systems. It is not, and that is the subject of the next section.

7 Refuting automatic field-equation nonredundancy

Session four's decisive finding is a negative one, and we present it as such: the tempting conjecture that a nonsingular curve with an on-curve non-base target always yields nonredundant field equations is *false*, with an infinite explicit counterexample family.

Theorem 12 (an infinite redundant Semaev family). **PROVED** Let $q \equiv 3 \pmod{4}$, $q \geq 7$, and $\mathbb{F}_{q^2} = \mathbb{F}_q[u]/(u^2 + 1)$. Choose $h \in \mathbb{F}_q^\times \setminus \{1, -1\}$ and set

$$\rho = (h + h^{-1})/2, \quad \sigma = (h^{-1} - h)/2, \quad a = -4\sigma^2. \quad (12)$$

Then $E : Y^2 = X^3 + aX$ is nonsingular, the point $R = (2u, 2\rho(1 - u))$ lies on E with non-base x -coordinate, and the two coordinates of $f_3(x, y, 2u)$ generate the same ideal as

$$H_0 = (xy - a)^2 - 4(x - y)^2, \quad H_1 = (x + y)(xy + a). \quad (13)$$

Both field equations $x^q - x, y^q - y$ are redundant: they lie in $C = (H_0, H_1)$.

Proof. Since -1 is a nonsquare (as $q \equiv 3 \pmod{4}$) and $\rho^2 - \sigma^2 = 1$, one has $\rho \neq 0$ and, by the restriction on h , $\sigma \neq 0$, so $a \neq 0$ and E is nonsingular. Both sides of the curve equation at R equal $-8\rho^2 u$. The branch $x + y = 0$ of C gives the four points $(z, -z)$ with $z \in \{\pm 2(1 + \rho), \pm 2(1 - \rho)\}$; the branch $xy = -a = 4\sigma^2$ gives two ordered roots of sum $4\sigma\rho$ (discriminant $16\sigma^4$), their swap, and both negatives. These eight points are distinct and lie in $\mathbb{F}_q^2$; a collision between branches would make -1 a square. The universal basis gives quotient dimension 8 (its sole specialization factor is $(m_1^2 - 4m_0)t_1^2/4 = -4 \neq 0$ here), so eight distinct zeros force $A_C \cong \mathbb{F}_q^8$; by the redundancy criterion the field equations belong to C . $\square$

Corollary 13 (the smallest concrete counterexample). **PROVED** Take $q = 7$, $h = 2$, so $\rho = 3$, $\sigma = 1$, $a = 3$. Over $\mathbb{F}_{49} = \mathbb{F}_7[u]/(u^2 + 1)$ use $E : Y^2 = X^3 + 3X$ and $R = (2u, 6 + u)$. Then both field-equation normal forms vanish and

$$(d_{\text{ff}}, d_{\text{reg}}, \text{sd}_{\text{grevlex}}) = (5, 7, 5). \quad (14)$$

Proof. The core generators are $G_0 = x^2y^2 + 3x^2 + 2xy + 3y^2 + 2$ and $G_1 = 3x^2y + 3xy^2 + 2x + 2y$, with reduced grevlex basis $\{xy^3 - 3x^2 + xy - 2, y^4 - 3y^2 + 2, x^3 + x + y^3 + y, x^2y + xy^2 + 3x + 3y\}$. Exact division gives zero normal forms for both $x^7 - x$ and $y^7 - y$. The closed degree-4 space has rank 4 and omits two basis elements; the degree-5 space contains all four. Hence $\text{sd} = 5$, strictly below $d_{\text{reg}} = 7$. The certificate `certify_quadratic_field_equations.py` checks

the symbolic family over $\mathbb{Z}[1/2, h, h^{-1}, u]/(u^2 + 1)$ and the concrete normal forms, quotient multiplication matrices, and closed spaces. $\square$

This is a counterexample *inside the actual on-curve Semaev family*, not merely inside an abstract family with the same top-degree shape. Its eight core zeros, split cleanly across the two branches, are shown in the right panel of Figure 2. The left panel is the exhaustive census that both locates these six redundant $q = 7$ systems and confirms their absence at $q = 5, 9$.

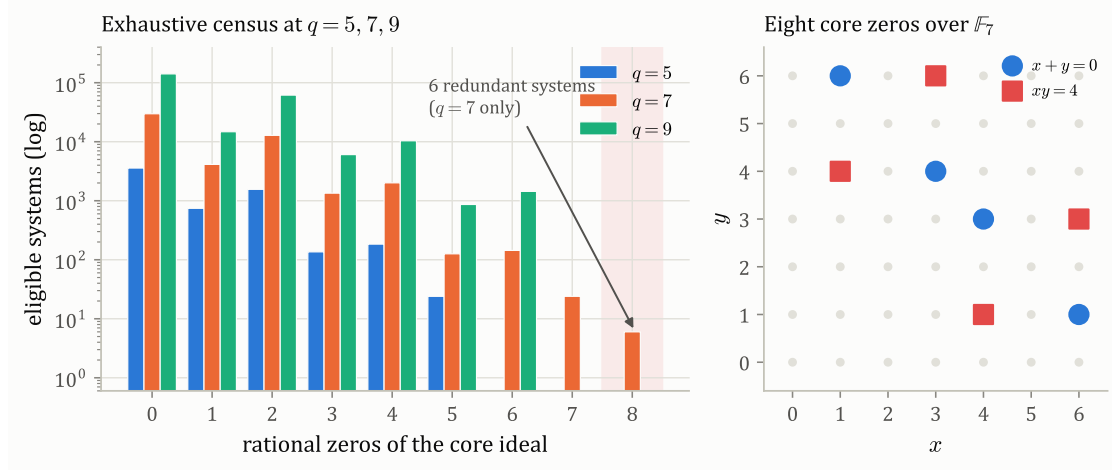


Figure 2: **EMPIRICAL: exhaustive $q=5,7,9$** Left: histogram of the number of rational core-ideal zeros over all eligible nonsingular curve/non-base-target systems; redundancy requires the maximum count of eight, achieved only by six systems at $q = 7$. Right: the eight base-field core zeros of the $q = 7$ counterexample, four on the branch $x + y = 0$ and four on $xy = 4$. Data: `search_quadratic_redundancy_20260716.json`, `certify_quadratic_field_equations_20260716.json`.

Consequently the unconditional statement is $\text{sd} \leq q$, and the equality $\text{sd} = q$ genuinely requires the nonredundancy hypothesis — which cannot be dropped. We flag this exactly as the research log does: a former six-prime solving-degree *conjecture* survives only as a *conditional* theorem.

8 Empirical validation

Every proved value is anchored by exact measurement, and the honest ranges, censored rows, and search counts are reported without smoothing.

8.1 The measured degree table

Figure 3 plots the two families that matter, and Table 1 tabulates the underlying rows. All values come from exact closed Macaulay spaces with SymPy-verified Gröbner-basis containment; a dash is a *censored* value, never an estimate.

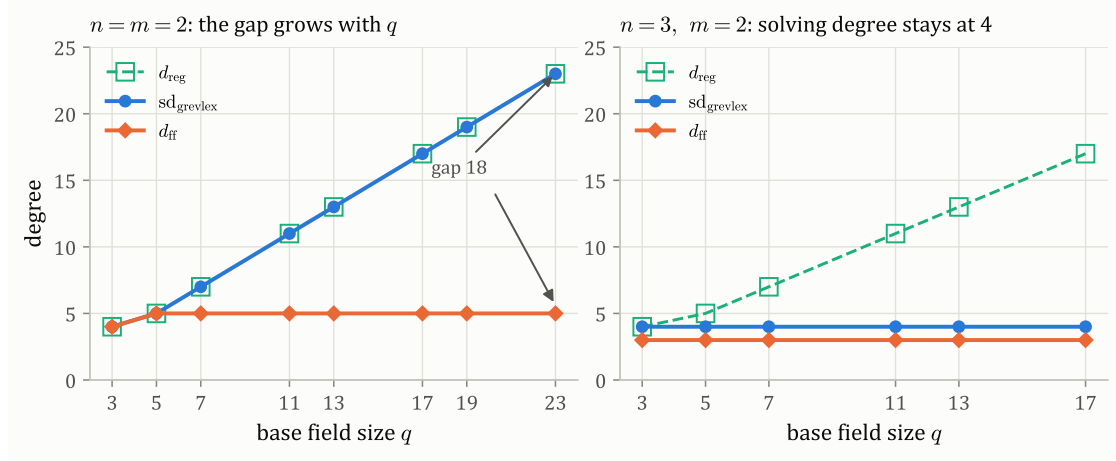


Figure 3: **EMPIRICAL: known-target rows, q up to 23** Left: $n = m = 2$. First fall stays at 5 while $sd = d_{\text{reg}} = q$, so the solving-minus-first-fall gap grows to 18 at $q = 23$. Right: $n = 3, m = 2$. Here sd stays at 4 while d_{reg} climbs with q — a live demonstration that d_{reg} and sd are different columns. Data: first_fall_vs_solving_20260701.csv.

Table 1: Known-target subset of first_fall_vs_solving_20260701.csv. The $n = m = 2$ block realizes the exact theorem; the $n = 3, m = 2$ block shows $sd = 4 < d_{\text{reg}} = q$; the last rows reach the resource boundary ($q = 5, m = 4$ is censored after the regularity stage at 60 s).

q	n	m	d_{ff}	d_{reg}	sd	$sd - d_{\text{ff}}$	root
3	2	2	4	4	4	0	✓
5	2	2	5	5	5	0	✓
7	2	2	5	7	7	2	✓
11	2	2	5	11	11	6	✓
13	2	2	5	13	13	8	✓
17	2	2	5	17	17	12	✓
19	2	2	5	19	19	14	✓
23	2	2	5	23	23	18	✓
3	3	2	3	4	4	1	✓
5	3	2	3	5	4	1	✓
7	3	2	3	7	4	1	✓
11	3	2	3	11	4	1	✓
13	3	2	3	13	4	1	✓
17	3	2	3	17	4	1	✓
5	2	3	12	12	12	0	✓
5	3	3	11	12	12	1	✓
3	2	4	8	8	8	0	✓
5	2	4	16	16	—	—	✓

Two independent separations appear in one table. In the $n = m = 2$ block, $sd - d_{\text{ff}}$ grows as $q - 5$; in the $n = 3, m = 2$ block, d_{reg} grows to q while sd is pinned at 4. The first refutes “first fall approximates solving”; the second refutes “regularity equals solving.” Both are exactly the confusions P1.3 targets.

8.2 Curve/target variation and exhaustive searches

Beyond the single known target, we varied curves and targets extensively. Table 2 collects the two kinds of evidence: a structured verified-root sample, and *exhaustive* enumeration at small q .

Table 2: Validation searches. The 397-case structured sample (quadratic_family_summary_20260709.csv) fixed a non-base coefficient of A and so *missed* the base-defined counterexample family — a selection bias the exhaustive $q = 7$ enumeration (search_quadratic_redundancy_20260716.json) exposed. The abstract trial (search_quadratic_counterexamples_20260709.csv) hits $c + g^2 = 0$, the very denominator pole excluded by the Semaev modulus.

Search	q	Eligible	Redundant	Outcome
Structured verified-root sample (fixed non-base A_1)	5	171	0	all $(5, q, q)$
	7	190	0	all $(5, q, q)$
	11	36	0	all $(5, q, q)$
Exhaustive curve/target	5	6,228	0	no redundant core
	7	50,376	6	six redundant systems
	9	236,160	0	no redundant core
Abstract top-shape trials	5	566	—	tuple $(3, 1, 4, 0, 1, 2, 0, 3)$: $\text{sd} = 6$

Three lessons are recorded. First, the exhaustive $q = 7$ search finds *all six* redundant systems and matches the infinite-family construction; the structured sampler found none because it fixed a nonzero extension coordinate of A , excluding the base-defined curves $Y^2 = X^3 + aX$ where redundancy occurs. Second, at $q = 9$ (the smallest non-prime odd prime power in scope) an exhaustive search over 236,160 eligible on-curve targets finds no redundant core, consistent with the family's $q \equiv 3 \pmod{4}$ hypothesis. Third, the abstract counterexample $(3, 1, 4, 0, 1, 2, 0, 3)$ with $\text{sd} = 6$ shows the top shape *alone* does not force $\text{sd} \leq q$: the nonvanishing of $c + g^2$ is essential, and it fails for that abstract tuple while holding for every genuine Semaev system.

9 Relation to prior bounds, obstruction, and limitations

The result sits precisely against the literature it draws on, and its boundaries are sharp.

Against the first-fall heuristic. Petit and Quisquater [4] assume $d_{\text{reg}} \approx d_{\text{ff}}$ for binary Weil systems; for our odd-characteristic quadratic family $d_{\text{reg}} = q$ while $d_{\text{ff}} = 5$, so the proxy is off by $q - 5$. This is not a contradiction of their asymptotic claim — their regime is binary with n large — but it is a concrete demonstration, on genuine Semaev systems, that the gap the heuristic assumes away can be arbitrarily large. Hodges, Petit, and Schlather [5] anticipated exactly this by labelling the closeness a heuristic; we make the failure quantitative.

Against general invariant-gap examples. Caminata and Gorla [7] already prove families with arbitrarily large gaps in either direction, and their Example 4.2 has $d_{\text{ff}} = 3$, $\text{sd} = q - 1$, $d_{\text{reg}} = 2q - 2$ even after field equations. Our contribution is *not* the existence of a gap in the abstract; it is the exact classification of a *specific, deterministic, odd-characteristic Semaev/Weil family* with verified roots. The local result is new only as an exact observation about this construction.

Against the solving-degree bounds we use. Caminata, Ceria, and Gorla [9] give general Weil-restriction solving-degree bounds via homogenized source regularity; Salizzoni [8] gives the closed-space bound we invoke. Converting the general bounds into a *sharp* explicit function of

q, n, m still needs control of the source-system regularity, which our experiments do not supply beyond $n = m = 2$. That is the honest obstruction.

Limitations, stated without hedging. The exact theorem holds only for $n = m = 2$ in odd characteristic. Automatic field-equation nonredundancy is *false*, so the equality half is conditional and the condition is not cosmetic. A bounded $n = 3, m = 2$ check over $q \in \{3, 5, 7, 11\}$ found core bases of degree ≤ 3 and field remainders of degree ≤ 2 , hinting at a low-degree mutant mechanism — but this is *empirical only*, not a symbolic $n = 3$ theorem. The $q = 5, m = 4$ cases were censored after the exact first-fall and regularity stages ($d_{\text{ff}} = d_{\text{reg}} = 16$) but before Gröbner-basis containment at a 60-second ceiling. And nothing here transfers to the binary $n \sim \log q$ regime studied by Kousidis and Wiemers [6]: their $d_{\text{ff}} \leq m^2 - m + 1$ bound and its asymptotic question live in a different characteristic and scaling, untouched by this odd-characteristic, fixed- n classification.

10 Conclusion

P1.3 asked for an unconditional, explicit degree bound on Weil-restricted Semaev systems that does not lean on the first-fall heuristic. For the smallest nontrivial family we delivered exactly that: $d_{\text{ff}} = 5$ and $d_{\text{reg}} = q$ are proved from the top-ideal shape, and $\text{sd}_{\text{grevlex}} \leq q$ is proved *unconditionally* through a constant-degree mutant family and Salizzoni’s regularity bound — a commutative-algebra argument, not a heuristic. Along the way the heuristic itself is shown to fail quantitatively for these systems: the solving-minus-first-fall gap grows as $q - 5$, reaching 18 at $q = 23$. The matching equality $\text{sd} = q$ is genuinely conditional on the field equations enlarging the core ideal, and we proved that this nonredundancy is *not* automatic by exhibiting an infinite on-curve counterexample family for every $q \equiv 3 \pmod{4}$, certified concretely at $q = 7$ and confirmed by exhaustive search. What remains open is the all-parameter bound and even a symbolic $n = 3, m = 2$ theorem; we record the promising bounded evidence for the latter as evidence, not as a claim.

Reproducibility

All computations are exact over prime fields and polynomial-basis extensions using Python 3.13.4 and SymPy 1.14.0; Sage, Singular, msolve, and Macaulay2 were unavailable in the working environment, and the closed-Macaulay solving-degree routine is a hand-built exact implementation. The measurement scripts are `measure_toy_degrees.py` (the four-degree separation), `sparse_weil.py` and `measure_semaev_stats.py` (Semaev expansion and Weil restriction), `measure_weil_degrees.py` (the canonical degree table), `measure_quadratic_variants.py` and `summarize_quadratic_variants.py` (curve/target variation), `certify_quadratic_family.py` (the localized-ring core certificate), `certify_quadratic_field_equations.py` (the infinite counterexample family and the $q = 7$ certificate), and `search_quadratic_redundancy.py` and `search_quadratic_counterexamples.py` (the exhaustive and abstract searches). Each degree row records the field model, curve, target, root check, engine/version, monomial order, resource ceilings, elapsed time, and last completed pipeline stage; censored rows are preserved as such and never extrapolated. Every mathematical claim above carries one of the epistemic tags `PROVED`, `CITED`, `CONDITIONAL: hypothesis`, or `EMPIRICAL: range` as used in the research log; untagged sentences are exposition, not claims.

References

- [1] I. Semaev, “Summation polynomials and the discrete logarithm problem on elliptic curves.” 2004.

- [2] P. Gaudry, "Index calculus for abelian varieties of small dimension and the elliptic curve discrete logarithm problem," *Journal of Symbolic Computation*, vol. 44, no. 12, pp. 1690–1702, 2009.
- [3] C. Diem, "On the discrete logarithm problem in elliptic curves," *Compositio Mathematica*, vol. 147, no. 1, pp. 75–104, 2011.
- [4] C. Petit and J.-J. Quisquater, "On polynomial systems arising from a Weil descent," in *ASIACRYPT 2012*, in LNCS, vol. 7658. 2012, pp. 451–466.
- [5] T. J. Hodges, C. Petit, and J. Schlather, "First fall degree and Weil descent," *Finite Fields and Their Applications*, vol. 30, pp. 155–177, 2014.
- [6] S. Kousidis and A. Wiemers, "On the first fall degree of summation polynomials," *Journal of Mathematical Cryptology*, vol. 13, no. 3–4, pp. 229–237, 2019.
- [7] A. Caminata and E. Gorla, "Solving degree, last fall degree, and related invariants," *Journal of Symbolic Computation*, vol. 114, pp. 322–335, 2023.
- [8] F. Salizzoni, "An upper bound for the solving degree in terms of the degree of regularity." 2023.
- [9] A. Caminata, M. Ceria, and E. Gorla, "The complexity of solving Weil restriction systems," *Journal of Algebra*, vol. 621, pp. 116–133, 2023.
- [10] A. Caminata and E. Gorla, "Solving multivariate polynomial systems and an invariant from commutative algebra," in *PQCrypto 2021*, in LNCS, vol. 12542. 2021, pp. 3–36.